

Lab 3: Architectural Pattern Selection & Justification

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System: Telemedicine Slot Booking & Prescription Portal (Problem Statement #11)

We chose Microservices Architecture for the Telemedicine Slot Booking & Prescription Portal System.

Architectural Choice

The system is decomposed into five independently deployable services: the Patient/Physician Portal (presentation), Booking Service, Video Room Service, Prescription Service, and Notification Service. Each service owns its own data and communicates with the others through well-defined provided/required interfaces.

Two Reasons for This Choice

1. Independent scaling under uneven load. Booking traffic spikes at the start of a consultation slot window, while the Video Room Service and Notification Service peak at the exact appointment time. As separate services, each can be scaled independently rather than over-provisioning a single monolith for the busiest module.

2. Fault isolation for a clinical system. If the Prescription Service or its external e-signature dependency fails (as happened with the blocked certificate dependency in Lab 2's sprint), patients can still browse specialties, book slots, and join consultations. A layered or client-server design would risk a single failure taking down the whole portal, which is unacceptable for a healthcare booking system.

Security Advantage

Each microservice enforces its own role-based access control at its interface boundary (NFR-002), so the Video Room Service can independently verify that only the booked patient and assigned physician reach a given room, and the Prescription Service can independently verify physician license credentials before signing - without depending on a shared trust boundary that a single compromised layer could bypass. This limits the blast radius of a breach to one service instead of the entire system.

Performance Benefit

Because services are decoupled, the latency-sensitive Video Room Service (real-time video signaling, NFR-001's TLS 1.3 encrypted session metadata) can be optimized and deployed on infrastructure suited to low-latency streaming, independent of the Booking Service's transactional database workload — avoiding the performance overhead a shared layered stack would impose on every request.